

Planned Large-Scale Experimental Protocol

SDAO validation plan; no future results are claimed in this appendix

Scope. This protocol extends the preliminary evaluation in §6 into a reproducible systems-validation plan. It specifies acquisition, controls, measurement, and statistical analysis before runs are performed.

Evaluation goals

1. Determine whether SDAO satisfies a stated filtered-recall target while reducing execution cost relative to fixed physical strategies.
2. Measure the accuracy and overhead of the selectivity, overlap, and cost estimates.
3. Separate cold-start, steady-state, and drift behavior.
4. Compare fairly against exact, native ANN, and PostgreSQL-based baselines.

Primary outcomes

Filtered Recall@K; result-shortfall rate; median/p95/p99 latency; QPS; planner, execution, and total latency; vectors scanned; index-build time and size; peak memory; cost-model error; and planner regret relative to an offline feasible oracle.

Datasets and Reproducible Predicate Workloads

Family	Role	Required record
SIFT1M	Correctness and calibration	Source, checksum, metric, split
SIFT10M	Scale, memory, build cost	Source, checksum, metric, split
DEEP1M	Cross-distribution validation	Source, checksum, metric, split
BEIR	Semantic workload heterogeneity	Dataset release, encoder, qrels, overlay

Predicate overlay

BEIR does not inherently provide a uniform SQL-filter workload. Use native metadata where available; otherwise generate deterministic, documented attributes from corpus properties such as document-length quantile, source collection, language, or fixed hash bucket. Never derive predicates from relevance labels. For each query/predicate pair, compute filtered exact top-K ground truth and separately report cases with fewer than K eligible documents.

Coverage matrix

Test equality, range, conjunction, and disjunction predicates across selectivity buckets <0.1%, 0.1--1%, 1--5%, 5--20%, 20--50%, and >50%. Include anti-correlated, approximately independent, and positively correlated predicate--vector regimes.

Controlled Platform and Measurement Contract

Reference server specification

Two AMD EPYC 9554 processors (64 physical cores each), 512 GiB DDR5 ECC memory, high-performance 3.84 TB NVMe storage, Ubuntu 24.04 LTS, and a pinned PostgreSQL/pgvector/hnswlib toolchain. The final artifact must record exact package versions, compiler flags, kernel version, NUMA policy, CPU governor, Docker image digest, and configuration files.

Timing decomposition

For every query, report $T_{total} = T_{plan} + T_{exec}$, where $T_{plan} = T_{features} + T_{estimator} + T_{cost} + T_{decision}$. Report warm-cache and cold-cache trials separately. Exclude evaluation-only exact-ground-truth construction from online query latency, but report that cost independently.

Load modes

Use closed-loop latency runs and open-loop QPS runs at controlled client concurrencies of 1, 8, 32, and 128. Pin processes to cores, perform at least 1,000 warm-up queries, and run no unrelated workload.

Baselines, Tuning, and Oracle Controls

Method	Purpose
Exact SQL-first / PRE	Exact quality and predicate-first control
pgvector exact	PostgreSQL execution baseline
pgvector HNSW / IVFFlat	Integrated approximate baselines
hnswlib POST	Native vector-first post-filter baseline
Filtered hnswlib GRAPH proxy	Native predicate-filtered traversal proxy
Forced SDAO strategies	PRE, POST, GRAPH, and ISECT action values
Static histogram selector	Estimator-free planner control
Offline feasible oracle	Per-query lower bound and regret reference

Tune all ANN parameters only on a disjoint tuning workload. Fix HNSW construction and search parameters, pgvector settings, and any oversampling cap before final tests. Include ACORN only when the authors' native artifact can be compiled and configured faithfully; otherwise report it as unavailable rather than treating filtered hnswlib as ACORN.

Multi-Seed Statistical Validation Pipeline

Run schedule

Use at least ten independent seeds. Vary and log data/query generation, index-construction order, estimator initialization, calibration split, and online feedback order. For each seed, execute the same ordered test trace for all paired methods. For online learning, evaluate each query before using its feedback and report cold-start separately from steady state.

Paired analysis

Retain one row per query execution. For baseline b , analyze paired differences $\Delta T(q) = T(\text{SDAO}, q) - T(b, q)$ and $\Delta R(q) = R(\text{SDAO}, q) - R(b, q)$. Report mean and median differences, 95% stratified bootstrap confidence intervals, effect sizes, recall-target violation rate, and oracle regret.

Hypothesis testing

Use a paired t-test only when paired-difference assumptions are defensible; otherwise use a two-sided Wilcoxon signed-rank test. Apply Holm correction across the pre-registered comparisons. Report sample size, test statistic, adjusted p-value, confidence interval, and effect size. Do not claim statistical significance until the planned measurements are collected.

Planner Decision Summary

Automated planner decision artifacts generated from latest runs

Target recall used: 0.95

Planner behavior: Strategies are filtered by estimated recall before selecting the lowest-cost feasible option.

Notes: This report is generated from the updated planner implementation and is intended to support the paper's feasibility discussion.

Planner Decisions by Strategy

Counts of queries for which each strategy was chosen



Recall@10 vs Latency (ms)

Per-query points colored by strategy and Pareto envelope

